

Elastic Load Balancer.

What ELB actually does

ELB automatically **distributes incoming traffic across multiple targets** (EC2, containers, IPs) so your app:

- doesn't crash under load
- stays highly available
- scales without manual intervention

If you're not using a load balancer in production, you're asking for downtime. Simple.

Types of Load Balancers.

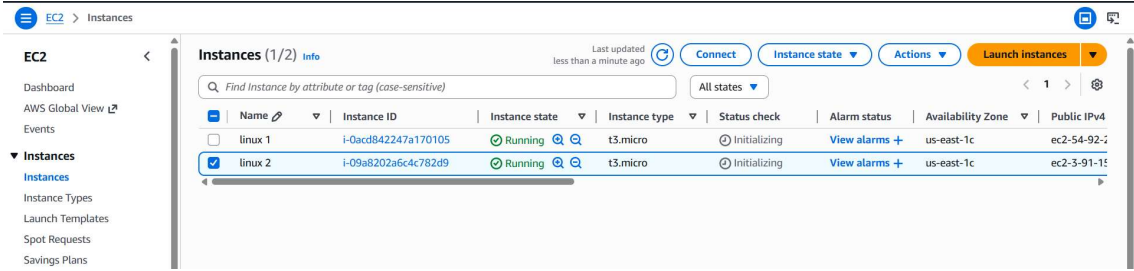
- **Application Load Balancer (ALB):** This operates at the application layer (Layer 7 of the OSI model) and is ideal for HTTP and HTTPS traffic.
 - **Key Features:** It provides advanced, content-based routing rules (e.g., based on URL path, host headers, HTTP methods, or query string parameters) to direct requests to different backend services.
 - **Best For:** Microservices architectures, container-based applications, and general web applications.
- **Network Load Balancer (NLB):** This operates at the transport layer (Layer 4) and is best suited for high-performance TCP, UDP, and TLS traffic.
 - **Key Features:** It offers extremely high performance, very low latency (~100 microseconds), and the ability to use static IP addresses. It also preserves the source IP of the client.
 - **Best For:** Extreme performance scenarios, workloads requiring static IPs, or non-HTTP/S protocols.
- **Gateway Load Balancer (GWLB):** This operates at the network layer (Layer 3) and is designed for deploying, scaling, and managing third-

party virtual appliances such as firewalls, intrusion detection systems, and other security or networking appliances.

- **Key Features:** It uses the GENEVE protocol to exchange application traffic and operates transparently within the network path.
- **Best For:** Centralized inspection and management of network traffic using virtual appliances.
- **Classic Load Balancer (CLB):** This is the previous generation of load balancers, still available but not recommended for new applications.
 - **Key Features:** It operates at both Layer 4 and Layer 7, providing basic load balancing across multiple EC2 instances. AWS encourages migration to ALB or NLB for enhanced capabilities.
 - **Best For:** Legacy applications that are still running on the EC2-Classic network.

I have created the Classic Load balancer with two instances.

Two Created EC2 Instances.



The screenshot displays the AWS Management Console's 'Instances' page. On the left, a navigation menu shows 'EC2' selected, with options for Dashboard, AWS Global View, Events, Instances (expanded), Instance Types, Launch Templates, Spot Requests, and Savings Plans. The main panel, titled 'Instances (1/2)', shows a table of two EC2 instances. Both instances are in a 'Running' state. The first instance is named 'linux 1' with ID 'i-0acd842247a170105'. The second instance is named 'linux 2' with ID 'i-09a8202a6c4c782d9'. Both are 't3.micro' instances in the 'us-east-1c' availability zone. The table includes columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. Action buttons like 'Connect', 'Instance state', 'Actions', and 'Launch instances' are visible at the top right of the table area.

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	linux 1	i-0acd842247a170105	Running	t3.micro	Initializing	View alarms +	us-east-1c	ec2-54-92-1...
☒	linux 2	i-09a8202a6c4c782d9	Running	t3.micro	Initializing	View alarms +	us-east-1c	ec2-3-91-1...

Configuration of Elastic Load Balancer – Classic .

Create Classic Load Balancer [Info](#)

The Classic Load Balancer distributes incoming application traffic across multiple EC2 instance targets in multiple Availability Zones. This increases the fault tolerance of your applications. Elastic Load Balancing detects unhealthy instances and routes traffic only to healthy instances.

► **How Classic Load Balancers work**

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

myclassic

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves Internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your network settings.

VPC [Info](#)
Select the virtual private cloud (VPC) for your targets. Only VPCs with an internet gateway are available for selection. The selected VPC cannot be changed after the load balancer is created. When selecting a VPC for your load balancer, ensure each subnet has a CIDR block with at least a /27 bitmask and at least 8 free IP addresses.

[Learn more](#)

Then we have to add instance to the Load Balancer.

Instances (2) Remove Add instances

You can add instances to register as targets of the load balancer. Alternatively, after your load balancer is created, you can add it to an Amazon EC2 Auto Scaling group to ensure you maintain the correct number of instances to handle the load for your application. For maximum fault tolerance, we recommend maintaining approximately equivalent numbers of instances in each Availability Zone.

Filter instances

☐	Instance ID	Name	State	Security groups	Zone	Public IPv4 address
☐	i-0acd842247a170105	linux 1	Running	launch-wizard-1	us-east-1c	54.92.212.245
☐	i-09a8202a6c4c782d9	linux 2	Running	launch-wizard-1	us-east-1c	3.91.154.206

After creation when we hit the load balancer using its DNS will give us output.

EC2 > Load balancers > myclassic

Capacity Manager [New](#)

Images
AMIs
AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups
Trust Stores

Auto Scaling
Auto Scaling Groups

Settings

Successfully created load balancer: **myclassic**

It might take a few minutes for your load balancer to be fully set up and ready to route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

myclassic

[Refresh](#) [Actions](#)

Details

Load balancer type
Classic

Status
0 of 2 instances in service

VPC
[vpc-0be33b50250021280](#)

Date created
March 20, 2026, 10:17 (UTC+05:30)

Scheme
Internet-facing

Hosted zone
Z35SXDOTRQ7X7K

Availability Zones

- [subnet-0c69ec4b1fefaac93](#) us-east-1e (use1-az3)
- [subnet-0618e0bb75502f409](#) us-east-1a (use1-az1)
- [subnet-013483e4615acb32f](#) us-east-1d (use1-az6)
- [subnet-07cb53f3c36994bc1](#) us-east-1c (use1-az4)
- [subnet-0ca4939a51840d7c4](#) us-east-1b (use1-az2)
- [subnet-07e40a724fbc9e7ed](#) us-east-1f (use1-az5)

DNS name [Info](#)
[myclassic-1224567581.us-east-1.elb.amazonaws.com](#) (A Record)

This Classic Load Balancer can be migrated to a next generation load balancer. Migration wizard uses your load balancer's current [Launch migration wizard](#)

Instance – 1

hello from ip-172-31-20-170.ec2.internal

Instance – 2

hello from ip-172-31-28-243.ec2.internal

I have created the Application Load balancer with two instances.

For creating Application Load Balancer we have to Create Target groups for the specified page.

The screenshot shows the 'Create target group' page in the AWS Management Console. The breadcrumb trail is 'EC2 > Target groups > Create target group'. On the left, a sidebar shows 'Step 3' and 'Review and create'. The main content area is titled 'Settings - immutable' and includes a warning: 'Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.'.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

- Instances** (selected): Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management. Suitable for: ALB, NLB, GWLB.
- IP addresses**: Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same Instance. Supports IPv6 targets. Suitable for: ALB, NLB, GWLB.
- Lambda function**: Supports load balancing to a single Lambda function. ALB required as traffic source. Suitable for: ALB.
- Application Load Balancer**: Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source. Suitable for: NLB.

Target group name
Name must be unique per Region per AWS account.
logintargets
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen, 1-32 total characters; Count: 11/32.

Protocol
Protocol for communication between the load balancer and targets.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4 (selected)
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

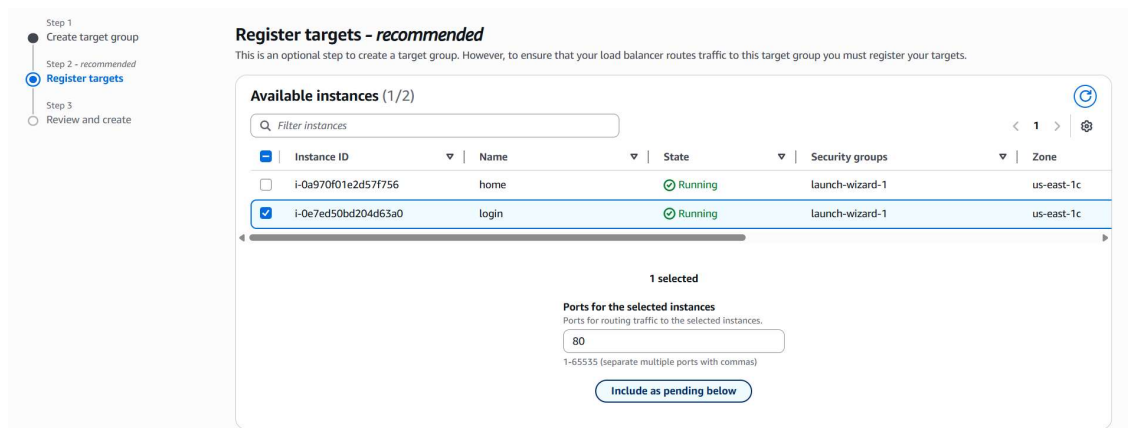
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

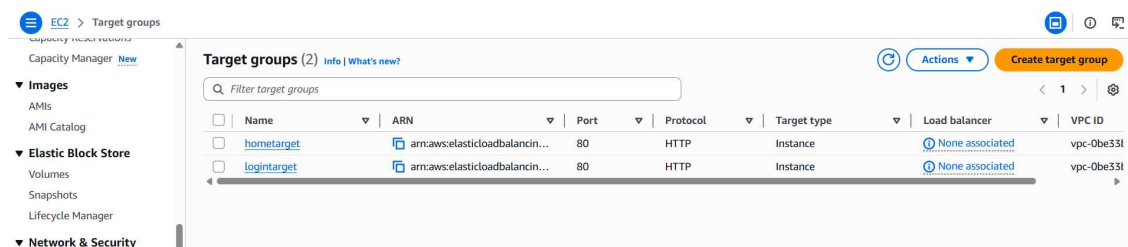
Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/login
Up to 1024 characters allowed.

► Advanced health check settings

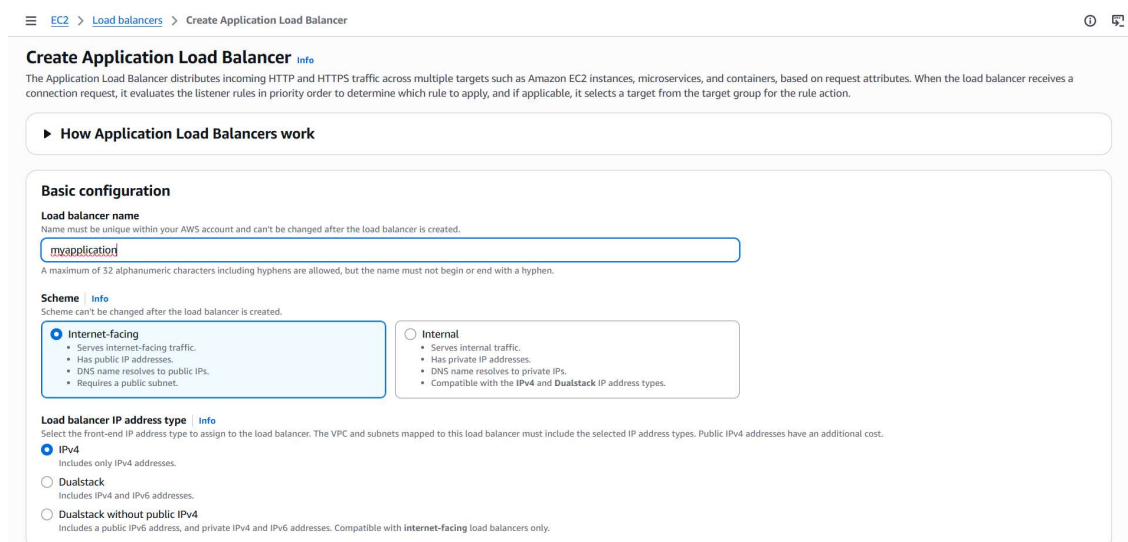
Then we have to assign EC2.



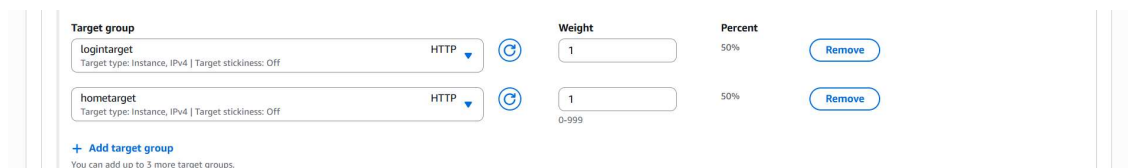
And I have Created Two Target Groups .



Then I have created Application Load Balancer .



Then I have selected Two Target Groups.



After the creation I have Two Separate page of hosting.

For Login Page.

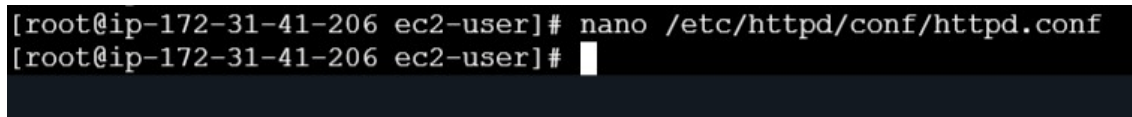


For Home page

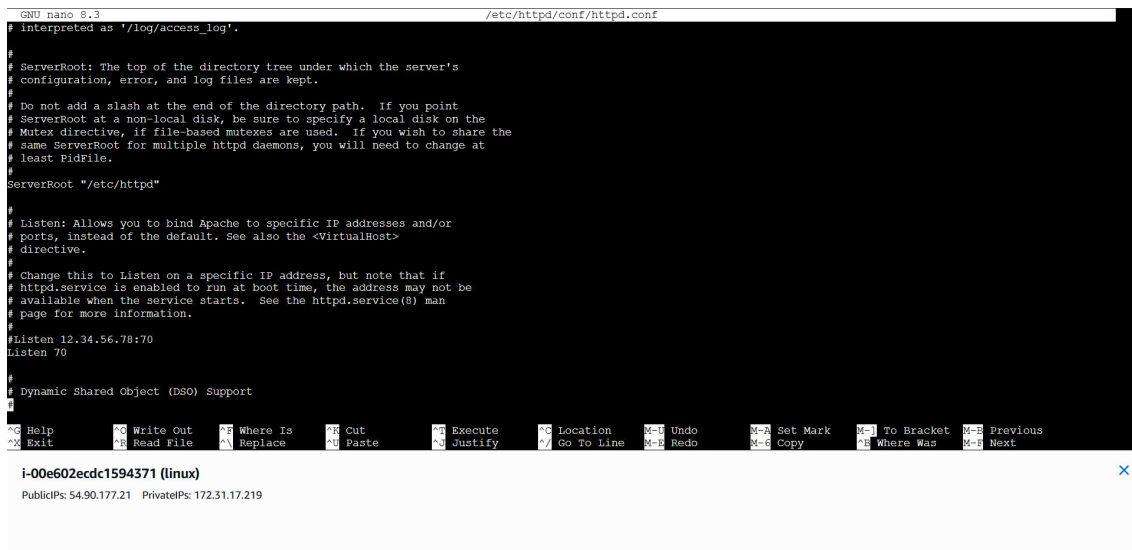


Changing the Httpd Web Server Port Number.

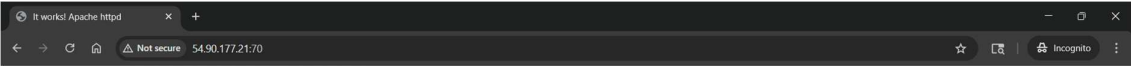
To change the default port in Apache (httpd), you need to modify the main configuration file(s) and then restart the service.



I have changed the listening port from 80 to 70.



After Changing the Port.



It works!